THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 8: **Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of all integers from 1 to 150 that are divisible by 4 is: 1

(a) 2808 **(b)** 2812 **(c)** 2816 **(d)** 2822

2. The sum of the first 14 terms of an AP is 147 and its first term is 30. The common difference is: 1

(a) -2 **(b)** -4 **(c)** -3 **(d)** -1

3. Three numbers in AP have sum 36 and product 1428 (with $d > 0$). The largest number is: 1

(a) 17 **(b)** 18 **(c)** 12 **(d)** 22

4. A town's population is 20000 and grows by 20% each year. After 2 years it is: 1

(a) 28800 **(b)** 29000 **(c)** 29200 **(d)** 28600

5. The first term of a GP is 2 and the 6th term is 486 (with $r > 0$). The common ratio is: 1

(a) 4 **(b)** 2 **(c)** 5 **(d)** 3

6. The nth term of an AP is $t_n = 5n - 1$. The sum of its first 11 terms is: 1

(a) 314 **(b)** 319 **(c)** 330 **(d)** 324

7. The 3rd term of an AP is 25 and the 8th term is 70. The 11th term is: 1

(a) 88 **(b)** 115 **(c)** 106 **(d)** 97

8. The sum of the first n terms of a sequence is $S_n = 4n^2 + 5n$. Its 7th term is: 1

(a) 53 **(b)** 61 **(c)** 65 **(d)** 57

9. **Assertion (A):** The sum of the first 10 natural numbers is 55. 1
Reason (R): The sum of the first n natural numbers is $n(n+1)$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The sequence $3, 6, 12, 18, \ldots$ is a geometric progression. 1
Reason (R): In a geometric progression the ratio of consecutive terms is constant.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Which term of the AP $11, 18, 25, \ldots$ is the first term to exceed 150? 2

12. If $S_n = 2n^2 + 5n$ is the sum of the first n terms of a sequence, find its first term and its 12th term. 2

13. The sum of the first 21 terms of an AP is 1218 and the first term is 8. Find d and the 21st term. 2

14. Which term of the AP $2, 11, 20, \ldots$ is the first term to exceed 301? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The sum of the 4th and 8th terms of an AP is 62 and its 7th term is 38. Find the AP's first term and the sum of its first 12 terms. 3

16. Find the sum of all natural numbers between 15 and 150 that leave remainder 1 when divided by 4. 3

17. Three positive numbers are in GP. Their product is 8000 and the sum of the first and third is 85. Find the numbers. 3

OR

The sum of the 5th and 11th terms of an AP is 32 and its 9th term is 14. Find the AP's first term and the sum of its first 16 terms. 3

18. Find the sum of all natural numbers between 30 and 250 that leave remainder 4 when divided by 9. 3

OR

Three positive numbers are in GP. Their product is 3375 and the sum of the first and third is 50. Find the numbers. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** An auditorium has 15 rows of seats. The first row has 20 seats and each next row has 2 seats more than the row in front of it. 5

(i) Find the common difference.

(ii) Find the number of seats in the 15th row.

(iii) Find the total number of seats in the auditorium.

(iv) Find the number of seats in the last 5 rows (11th to 15th rows).

20. A contractor pays a penalty for delay: ₹150 on day 1, ₹180 on day 2, ₹210 on day 3 and so on. If the work is 28 days late, find the penalty on the last day and the total penalty. 5

OR

A ball is dropped from 256 m and rebounds to 3/4 of the previous height. Show that the rebound heights form a GP and find the 3rd rebound height. 5

END OF QUESTION PAPER — Total: 40 Marks